A variable, circularly polarised 144 MHz source on the A2V star 50 Cassiopeiae

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Abstract

An A star has no convective envelope to drive a corona, and the A stars LOFAR has detected at 144 MHz, two magnetic chemically peculiar stars, show circular polarisation well below 60 per cent. In the per-pointing Stokes V images of the third data release of the LOFAR Two-metre Sky Survey, 50 Cas (HD 12216, A2V, 48 pc, $V = 3.95$) is the one A star within 50 pc of 129 measured that carries a polarised source. Three epochs cover it. On 2020 September 28 and 2022 April 10 a source of 1.7 and 2.6 mJy beam^{-1} stands on the star with Stokes V about 60 per cent of its peak, and the two epochs give $+1.13 \pm 0.14$ mJy beam^{-1} at the star's position (7.9σ); on 2022 January 6 the star's position holds $+0.08 \pm 0.15$ mJy beam^{-1}, so the polarised flux changes between epochs ($\chi^2 = 30$ for 2 degrees of freedom). At 48 pc the April source has a spectral luminosity of 7×10^{15} erg s^{-1} Hz^{-1}, and confined to the star's own radius it would have a brightness temperature of 7×10^{11} K. TESS's light curves of nine sectors from 2019 to 2024 hold a stable double wave with unequal maxima at 1.596538 d, semi-amplitudes of 2.88 mmag at half the period and 1.54 mmag at the period, which Hipparcos recorded at its first harmonic; at the star's radius of 2.98 $R_\odot$ the period gives an equatorial speed of 94 km s^{-1}, against a measured $v \sin i$ of 85–91 km s^{-1}. Coudé spectra of 1992 hold a companion on that period below about 0.17 $M_\odot$, which cannot make the light curve's term at the period. The star reads as a magnetic A star with surface spots turning in 1.6 days, whose electron-cyclotron emission is beamed across Earth at some rotational phases; no measurement of its field exists, and one would settle it.

Keywords: Radio continuum: stars, Stars: individual: 50 Cassiopeiae, Stars: chemically peculiar, Stars: magnetic field, Stars: rotation, Masers

1. Introduction

Main-sequence A stars lack the convective envelopes that power the coronae of cooler stars, and their radio emission comes from the magnetic chemically peculiar stars, whose large-scale dipolar fields trap electrons in a magnetosphere. These emit gyrosynchrotron radiation at gigahertz frequencies, with luminosities that scale with the field and the rotation [1], and some emit coherent pulses by the electron-cyclotron maser, strongly circularly polarised and beamed, which arrive at the rotational phases where the line of sight crosses the magnetic equator, as first seen in CU Vir [2]. At 144 MHz, LOFAR detected two magnetic chemically peculiar stars, BP Boo and α^2 CVn, in total intensity, with circular polarisation below 60 per cent [3]. The strongly circularly polarised stellar sources at these frequencies are active M dwarfs [4] and RS CVn binaries [5], catalogued over a quarter of the northern sky by the circularly polarised survey of LOFAR's second data release [6].

The third data release of the LOFAR Two-metre Sky Survey (LoTSS-DR3) covers 88 per cent of the northern sky and publishes a Stokes V image of each pointing's epoch [7]. Read at the

nearest stars, those images show the flare stars and active binaries already known, and one A star. This note reports that star, 50 Cas, and what the archives give for its nature.

1.1. Contributions

1. A circularly polarised, variable 144 MHz source on 50 Cas in three epochs of LoTSS-DR3, the only one among 129 A and early-F stars within 50 pc.
2. The star's 1.5965 d cycle in nine TESS sectors and its consistency with the measured rotational broadening, and the limit that spectra of 1992 set on a companion on that cycle.

2. 50 Cassiopeiae

50 Cas (HD 12216, HR 580, HIP 9598) is a naked-eye A2V star. Gaia DR3 [8] gives a parallax of 20.84 ± 0.19 mas (48.0 pc), $G = 3.95$, $G_{\rm BP} - G_{\rm RP} = 0.06$, a renormalised unit weight error of 3.2, which is common at $G \approx 4$, no non-single-star solution and no radial velocity, and from its spectrophotometry an effective temperature of 9060 K, a radius of $2.98\,R_\odot$, a luminosity of $54\,L_\odot$, a mass of $2.4\,M_\odot$ and an age of 0.5 Gyr; the debris-disc survey of Rieke et al. gives it 320 Myr [9]. Its projected rotational velocity is $85\,{\rm km\,s^{-1}}$ [10] and 90–91 $\rm km\,s^{-1}$ [11, 12]. Three classifications of 1954–1960 collected by SIMBAD read A1 V SB. The catalogue of Ap, HgMn and Am stars [13] does not list it. It was among the targets of a magnetic survey of the bright northern main-sequence stars [14], and the compilation of averaged effective fields of Bychkov et al. [15], which carries that survey's measurements of 28 stars, holds none of 50 Cas. The Hipparcos–Gaia proper-motion anomaly stands at a signal-to-noise ratio of 1.5–1.7 [16]. Koen and Eyer found a period in its Hipparcos photometry, $1.25273\,{\rm d^{-1}}$ at 3 mmag [17]. IRAS's 25 μm flux reads 1.26 times the photosphere [9], listed among the less certain excess stars [18], while AllWISE's W3 and W4, 3.91 and 3.86 mag against 2MASS $K_s = 3.92$, allow no excess above about five per cent at 12 and 22 μm. No other Gaia source lies within 24″; the nearest, at 33″ and 51″, are 10.5 mag fainter. The star is a target of Breakthrough Listen's nearby-star survey [19].

3. Data and method

LoTSS-DR3. Each DR3 pointing is an 8 h observation of one epoch. The ASTRON VO's table of DR3 pointings gives the midpoint of each observation, and the SURF Data Repository holds each pointing's products as tar archives, from which the 6″ dirty Stokes I and V images and the 6″ restored Stokes I image with astrometric corrections were read by HTTP byte range, a patch around the star alone, and put on DR3's flux scale by the factor the release gives for the pointing. The search that found 50 Cas measured Stokes V at the 246 Hipparcos stars within 50 pc with $-0.05 < B-V < 0.32$ north of $-1°$ in their two nearest pointings, each star carried to the pointing's epoch, its noise from the robust scatter of the annulus 12″–30″ out, positions 22.5″ away serving as the null; 61 of the 190 measured carry a placeholder colour or an absolute magnitude fainter than 3.5 and are later stars, leaving 129 A and early-F stars at a median 5σ limit of 0.54 mJy. For 50 Cas every pointing within 3° was read, three of which cover it.

TESS. The two-minute PDCSAP light curves of sectors 18, 19, 25, 52, 58, 59, 78, 79 and 86, 141552 cadences of quality zero from 2019 to 2024, in mmag about each sector's median, were fitted with a fundamental frequency and its first three harmonics and one offset per sector over a fine grid of frequencies, the frequency's error from the curvature of χ^2 with the residual scatter as the noise. The ephemeris for phases is $0.62635569\,{\rm d^{-1}}$ with phase zero at BJD 2457000.

Table 1: 50 Cas in the per-pointing images of LoTSS-DR3: the pointing, the date of the observation's midpoint, the star's distance from the pointing centre, DR3's flux-scale factor, the phase range of the 8 h observation on the TESS ephemeris, the dirty Stokes I peak within 4.5″ of the star and its offset, and Stokes V at that peak and at the star's position, all on DR3's flux scale.

Pointing	Date	Offset (°)	Scale	Phase	I (mJy beam^{-1})	(″)	$V_{\rm peak}$	$V_{\rm star}$
P037+71	2020 Sep 28	2.27	1.307	0.64–0.85	1.73 ± 0.29	3.4	+1.16	$+0.77 \pm 0.22$
P032+74	2022 Jan 6	1.49	0.958	0.75–0.96	0.30 ± 0.17	2.1	0.00	$+0.08 \pm 0.15$
P029+71	2022 Apr 10	1.20	1.084	0.40–0.61	2.58 ± 0.28	1.5	+1.53	$+1.38 \pm 0.19$

DAO spectra. The Canadian Astronomy Data Centre holds 19 McKellar coudé spectra of 50 Cas from the Dominion Astrophysical Observatory's 1.2 m telescope, from 1990, 1992, 2000 and 2006. The 13 of 1992 December 18 cover 3.12 h on one setup at 2.4 Å mm^{-1} near 4486–4558 Å, with 39 flats and Th–Ar arcs interleaved. Each exposure was divided by the median flat and cross-correlated with the series mean, and the arcs' drift of 0.8 km s^{-1} through the night was removed.

4. Results

4.1. The 144 MHz source

50 Cas is the only one of the 129 A and early-F stars above 3σ in Stokes V, at $+5.4\sigma$ over its two nearest pointings. Table 1 gives the three epochs. On 2022 April 10 the dirty images hold a Stokes I peak of 2.58 ± 0.28 mJy beam^{-1} 1.5″ from the star and Stokes V of $+1.53 \pm 0.19$ mJy beam^{-1} there, a fraction of 0.59; the restored image with astrometric corrections places the Stokes I peak 0.07″ from the star at a signal-to-noise ratio of 16.5. On 2020 September 28 the source stands at 1.73 ± 0.29 mJy beam^{-1} with Stokes V of $+1.16 \pm 0.22$ mJy beam^{-1} at the peak, a fraction of 0.67. On 2022 January 6, between them, nothing stands at the star: Stokes I reaches 1.8σ and Stokes V at the star's position is $+0.08 \pm 0.15$ mJy beam^{-1}. Read at the star's position, the two epochs with the source give $+1.13 \pm 0.14$ mJy beam^{-1} (7.9σ), and the three epochs are inconsistent with one value, $\chi^2 = 30.3$ for 2 degrees of freedom.

At 48.0 pc the April flux density corresponds to a spectral luminosity of 7.1×10^{15} erg s^{-1} Hz^{-1}, 3.8×10^{15} erg s^{-1} Hz^{-1} of it circularly polarised. Emission confined to a region of the star's radius would have a brightness temperature of 7×10^{11} K, and one of five stellar radii 3×10^{10} K. Nothing lies within 60″ in the ROSAT All-Sky Survey catalogues, nor in NVSS or TGSS at the star, and the four VLASS quick-look epochs of 2017–2026 hold nothing above 1.8σ at 3 GHz, 5σ limits of 0.8–1.1 mJy or 2×10^{15} to 3×10^{15} erg s^{-1} Hz^{-1}.

4.2. The 1.6-day cycle

The TESS fit gives $f = 0.62635527 \pm 0.00000006$ d^{-1}, a period of 1.596538 d, with semi-amplitudes of 1.54 mmag at f, 2.88 mmag at $2f$ and 0.11 and 0.07 mmag at $3f$ and $4f$, and residuals of 0.40 mmag per two-minute cadence. Each sector alone gives 1.45–1.59 mmag at f and 2.81–2.96 mmag at $2f$, so the curve has held its shape for five years. Folded (Figure 1), it is a double wave with unequal maxima, 4.3 mmag above the mean at phase 0.27 and 1.4 mmag at 0.80, and unequal minima, 3.7 mmag below at 0.53 and 2.1 mmag at 0.02. The frequency Hipparcos found is $2f$. No Gaia source within 90″ is brighter than $G = 14.5$, and none can carry a modulation of 0.7 per cent of a fourth-magnitude star's light, so the cycle is 50 Cas's own.

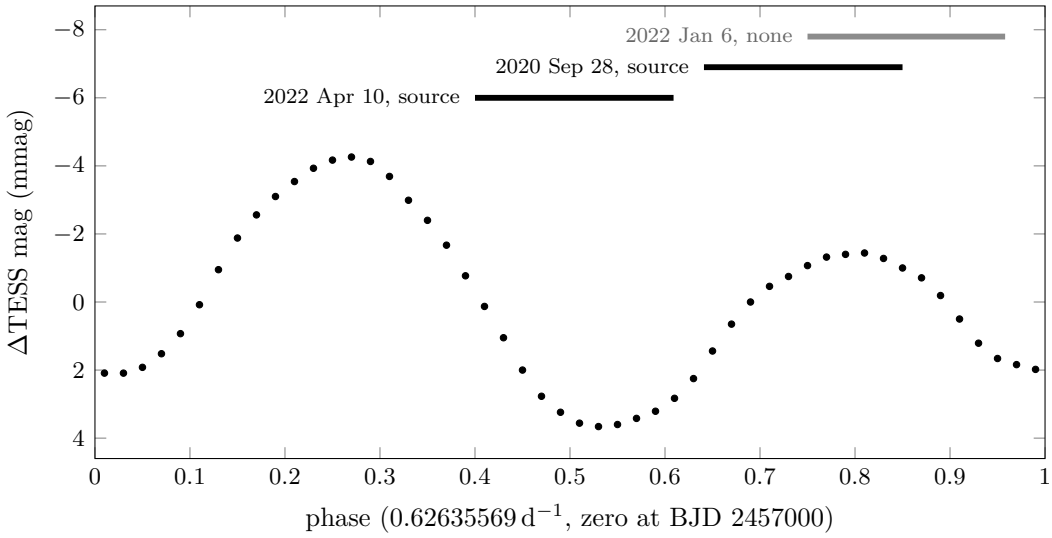


Figure 1: The TESS light curve of 50 Cas over nine sectors of 2019–2024, folded on the ephemeris and binned in 50 phase bins (errors smaller than the points), brighter upward; the bars mark the phases each 8 h LoTSS-DR3 observation covered, black where the polarised source was present and grey where it was not.

At the star's radius of 2.98 $R_\odot$, turning once in 1.5965 d moves the equator at 94 km s^{-1}, and the measured $v \sin i$ of 85–91 km s^{-1} then puts the inclination above about 65°, the star seen nearly equator-on. Turning once in half the period would need 188 km s^{-1} at the equator and an inclination near 30°, and would leave no light at f itself.

4.3. A companion on the cycle

The 1992 series runs from phase 0.297 to 0.379, and across its 3.12 h the spectrum moves by $+1.0 \pm 0.4$ km s^{-1} (a slope of $+0.31 \pm 0.12$ km s^{-1} h^{-1} with a scatter of 0.38 km s^{-1}). If the cycle were an orbit with conjunctions at the light minima, a companion would change the A star's velocity by 0.15 to 0.21 times its semi-amplitude K across those phases, so K lies below about 15 km s^{-1} (2σ) and the companion below about 0.17 $M_\odot$ at an inclination near 90°. Such a companion can make the 2.88 mmag ellipsoidal term at $2f$, but not the 1.54 mmag term at f: an M dwarf 8 $R_\odot$ from a 54 $L_\odot$ star reflects about 0.15 mmag and Doppler beaming at 15 km s^{-1} gives about 0.2 mmag. A companion of A type would shift the lines by tens of km s^{-1} across the series and double them, and does neither. The two spectra of 2000 September, at phases 0.86 and 0.57, differ by 5.5 ± 2 km s^{-1} with the sign along the pixel axis undetermined, an orbit's K of about 7 km s^{-1} or the line-profile change that surface spots make.

5. Discussion

The polarised fraction, about 60 per cent, and the brightness temperature, 10^{10} to 10^{12} K for regions of one to five stellar radii, point to coherent emission, and the electron-cyclotron maser is the coherent process that a magnetised star sustains. Emission at the fundamental of the gyrofrequency at 144 MHz needs about 51 G where it forms, which a dipole of a kilogauss at the pole holds at about 2.7 stellar radii.

Two readings fit what the archives give. In the first, 50 Cas is a magnetic A star: the light curve, a double wave with unequal maxima that has kept its shape for five years, is that of abundance spots fixed on the surface of a magnetic chemically peculiar star turning nearly equator-on, and the radio emission is its magnetosphere's electron-cyclotron maser, beamed so that Earth sees it at some phases and not others, as CU Vir's pulses are [2]. The three epochs fit a pulse near phase 0.62 lasting about a tenth of a cycle, at the end of the April observation and the start of the September one and outside the January one, and they fit as well an emission that changes from year to year. The two LOFAR detections of magnetic chemically peculiar stars at 144 MHz are in total intensity, with circular polarisation below 60 per cent [3], so this would be the first of their class seen there by its coherent emission.

In the second reading the radio source is an unseen companion below $0.17\,M_\odot$: a late M dwarf held in a 1.6 d orbit would rotate fast enough to be active, and LOFAR hears active M dwarfs and RS CVn binaries by their coherent, strongly polarised emission [4, 5]. The 1.54 mmag term at f still needs an inhomogeneous surface on the A star, which neither reflection nor beaming supplies, so the light curve speaks for the A star's own spots under either reading. A measurement of the longitudinal field by spectropolarimetry would decide between them, a field of hundreds of gauss marking the star as magnetic; a high-resolution spectrum would show whether its abundances are peculiar, and time-resolved 144 MHz data, which the DR3 visibilities hold, would show whether the emission comes in pulses at fixed phases.

6. Related work

Radio emission from magnetic chemically peculiar stars has been studied at gigahertz frequencies for decades, with a scaling of the incoherent luminosity with field and rotation [1] and coherent pulses since CU Vir [2]; at 144 MHz the LoTSS second data release yielded two detections in total intensity [3], and its circularly polarised catalogue covered areas that do not include 50 Cas [6]. The photometric period of 50 Cas was found in the Hipparcos photometry [17], and its rotational broadening is measured in the surveys of A-star rotation [10, 11, 12]. No earlier radio detection of 50 Cas, and no measurement of its magnetic field, is found in the literature.

7. Conclusion

50 Cas, an A2V star at 48 pc, carries a 144 MHz source with circular polarisation of about 60 per cent in two of the three LoTSS-DR3 epochs that cover it, absent in the third, at a luminosity of $7 \times 10^{15}\,\mathrm{erg\,s^{-1}\,Hz^{-1}}$ and a brightness temperature that requires coherent emission. The star turns in 1.5965 d nearly equator-on, its light a stable double wave with unequal maxima, and any companion on that cycle is below about $0.17\,M_\odot$. It reads as a magnetic A star whose electron-cyclotron emission is beamed across Earth at some rotational phases, with an active low-mass companion as the alternative. A spectropolarimetric measurement of its field would settle which.

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